

Fairweather: workflow note

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Live site: <https://fairweather-plum.vercel.app>

Repo: <https://github.com/debugagu/fairweather>

Who it advises. One person on foot, outdoors for most of the day, away from shelter. A day hiker. Every number below follows from that choice. Thirty two degrees is a pleasant afternoon if you can step into a shop and a real problem four hours into a walk, so thresholds tuned to cover both would be too vague to act on.

How it works. The typed city goes to Open-Meteo's geocoding endpoint, which returns up to five matches. A single match is used and named on screen; two or more get a screen of their own showing region, country, population and coordinates, because silently taking the first result guesses on the user's behalf. The chosen coordinates and dates then fetch twelve daily fields and three hourly series with `timezone=auto`, so days break at local midnight. If a climbing height was given, every figure is re-derived for that height before anything is judged. Seven concerns are then scored independently from 0 to 3: rain, heat, cold, wind, UV, daylight and the snow line. The highest writes the day's verdict, and ties break on heat, wind, cold, snow line, rain, UV, daylight, ordered by what actually causes harm. Concerns that did not write the verdict still feed one deduplicated packing list covering the whole trip.

Thresholds. Rain needs probability and volume together, because a 70% chance of half a millimetre is nothing while 12mm at 55% soaks through anything that is not a proper shell: 35% or 1mm to mention it, 60% with 2mm to plan around, 70% with 10mm to avoid. Heat reads apparent temperature rather than the thermometer, since humidity decides how hard a day of walking feels: 28, 32, 38 degrees. Cold reads the apparent minimum, which is what you meet at dawn and at every stop: 7, 2 and -5 degrees. Wind uses gusts rather than the sustained average, because the gust is what decides whether a route is sensible: 25, 40 and 60 km/h. UV runs 6, 8 and 11, and matters more here than for most people because there is no shade to step into. Daylight is capped at level 1, since short days never make weather bad, they only limit how far you can go.

Altitude. A forecast describes the town, and a hiker spends the day above it. Chamonix sits at 1,060m. Enter 2,800m and the same week turns from "all four days are walkable" into "only one of the four is straightforward". Temperature moves at 6.5°C per 1,000m and UV at 10%. Wind is deliberately left alone, because a ridge is windier but no honest figure comes from a single elevation number, and the page says so on screen. The snow line is fetched rather than estimated, from Open-Meteo's hourly freezing level.

One thing I left out. The hourly chart. The app fetches an hourly series and draws none of it. The same data becomes one sentence instead: "the rain clears around 13:00, so there is a good afternoon in this if you start late." Cloud cover is drawn as an icon but never scored, because an overcast 19 degree day is excellent walking.

What I would build next. Wind that knows about terrain. Altitude adjusts temperature and UV, but wind is still a valley reading, and it is the signal most likely to turn a hard day into a dangerous one.

Built as plain HTML, CSS and JavaScript with no framework and no build step, deployed on Vercel, with AI assistance from Claude Code for the markup and the plumbing around the two API calls.

For the full reasoning behind every threshold, the five interface states and the writing, the complete note is on the site at <https://fairweather-plum.vercel.app/note>